Fiscal Year:	FY 2013	**Task Last Updated:**	FY 03/29/2013
PI Name:	Oates, Chris Ph.D.		
Project Title:	Development/Evaluation of a 10^-17 Yb Optical Clock for Space Applications		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	80305-3337	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Fundamental Physics NNH11ZTT001N
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No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Diddams, Scott (National Institute of Standards & Technology) Ludlow, Andrew (National Institute of Standards & Technology)		
Grant/Contract No.:			
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Task Description:	The primary goal of ESA's SOC program is to develop a high-performance optical frequency standard (clock) for operation in space with 1 x 10^-17 frequency uncertainty. NIST proposes to leverage its existing research on world-leading Yb lattice optical clocks to: (1) demonstrate a laboratory-based Yb lattice optical clock with frequency uncertainty of 1 x 10^-17 or better; (2) stabilize a space-friendly fiber frequency comb to the Yb clock signal and (3) develop a preliminary design for space-qualification by providing clock subsystem specifications. The proposed program will provide ESA with detailed information to make the most informed decisions on use of one or more optical frequency standards for the flight portion of the SOC program. Currently, ESA has identified the Sr and Yb lattice optical clocks as the most promising technologies. A Yb lattice clock has significant potential advantages over Sr in that Yb clocks can operate with solid state lasers much more likely to be practical in space operations. Several European		

Task Description:	groups are presently evaluating Sr's space prospects and Sr has thus received significant attention. For Yb, however, the most mature and high performance lattice clock resides in our lab at NIST. This Yb clock currently performs at least as well as the world's best Sr clocks, approximately 10^-16 frequency uncertainty in all cases, and continues to rapidly improve. Thus, we are uniquely qualified to play a vital role in the SOC project. For a modest investment, the proposed research program will ensure ESA has the best information and options to select the optical clock technology to meet SOC program goals, ensuring fully informed decisions affecting a much larger financial and scientific investment in the SOC program. Furthermore, by demonstrating a 10^-17 level Yb optical lattice clock, our proposed effort will achieve significant milestones currently outlined in the SOC program.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	We have made excellent progress on our NASA Fundamental Physics project in Year 1 of the program. The overall goal for our research in this program is to demonstrate the suitability of an Yb optical lattice clock for applications in space, in terms of clock performance (10^-17 uncertainty) and technical suitability. The specific goal for the first year of our research was to prepare a second version of the Yb lattice clock, which will then be used to attack the most challenging goal for this research, namely, clear demonstration of 10^-17 fractional frequency performance for the Yb lattice clock. The main challenges in the development of a second Yb system are the development of the multiple laser systems and the generation of ultra-narrow (< 5 Hz full-width half-maximum) spectroscopic signals. We happily report that we now have a fully operational second Yb lattice clock, which is performing at a level similar to the original, with a few improvements added to it as well (e.g., improved vacuum system and more robust blue trapping laser system). We have trapped 5-10 thousand atoms in the optical lattice and have resolved spectroscopic signals for the clock signal with a linewidth of 4 Hz, which we have used to stabilize our pre-stabilized clock laser. The key advantage provided by having a second Yb system is that we will be able to use the two systems to perform detailed evaluations of the key remaining effects that could undesirably shift the Yb clock frequency. We have already performed initial comparisons to evaluate the stability of the difference between the two clock frequencies. This is critical because the stability tells how high a clock precision can be reached for a given averaging time – in other words, it tells us how long we will need to average our clock signals to reach say, the desired 10^-17 uncertainty level. In this respect, our initial measurements are extremely encouraging. First, we have performed such measurements day after day, demonstrating that the new system is really ready for serious metrology. But the really exciting thing is that we have seen the signals average down to below a fractional uncertainty of 10^-17 in only half an hour of running time, an unprecedented level of clock stability. Furthermore, day-to-day these signals seem quite robust, indicating that we may not have any large unknown effects lurking in these systems. These results put us in a good position for the focus of Year 2 of the performance plan, which will be a detailed evaluation of the frequency shifts caused by the optical lattice that is used to trap our clock atoms (we note that during Year 1 we also published an article in Physical Review A, verifying reduction of a second frequency shifting effect, one due to blackbody radiation, at the 10^-17 level). In the remainder of Year 1 we will install an optical build-up cavity on the second Yb clock system. This will allow us to (temporarily) increase the lattice light intensity so as to amplify the shifts under investigation. We have constructed a test version of the build-up cavity using a mock vacuum system, and its performance met the required specifications (> 50x power build-up), so it should be ready for installation in the coming months.
Bibliography Type:	Description: (Last Updated: 04/05/2023)
Articles in Peer-reviewed Journals	Beloy K, Sherman JA, Lemke ND, Hinkley N, Oates CW, Ludlow AD. "Determination of the 5d6s 3D1 state lifetime and blackbody-radiation clock shift in Yb." Physical Review A. 2012 Nov;86(5):051404(R). http://dx.doi.org/10.1103/PhysRevA.86.051404 , Nov-2012
Papers from Meeting Proceedings	Sherman JA, Lemke ND, Hinkley N, Fox RW, Ludlow AD, Oates CW. "Polarizability of an optical lattice clock at 20 ppm." 2012 IEEE Conference on Precision Electromagnetic Measurements, Washington, D.C., July 1-6, 2012. IEEE Conference on Precision Electromagnetic Measurements (CPEM) 2012 Conference Digest, p. 274-275. http://dx.doi.org/10.1109/CPEM.2012.6250908 , Jul-2012